Finding Thurstone: modeling Comparative Judgment data with `R` (and `Stan`)

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Abstract

The Classical Comparative Judgment (CCJ) analysis has become the standard approach for analyzing CJ data across multiple fields. Researchers favor it because it offers a simple approach for measuring traits and conducting analyses. Three features underpin its simplicity: it relies on restricted Thurstone-based models to estimate latent traits, it gives limited consideration to multiple assessment design features, and it separates trait measurement from analysis. Recent studies, however, question whether these modeling assumptions hold in contemporary CJ applications and whether the limited consideration of relevant design features, together with the separation of measurement and analysis, may bias estimates and reduce the precision of inferences.

To address these concerns, Rivera et al. (2025) proposed the Information-Theoretical Comparative Judgment (ITCJ) analysis, an umbrella analytical workflow designed to accommodate the complexity observed in applied CJ settings. The approach enables researchers to develop models tailored to the assumed CJ data-generating process, thereby reducing reliance on simplifying assumptions. In addition, by integrating measurement, analysis, and relevant assessment design features within a single framework, the approach facilitates coherent estimation and inference. However, these potential advantages require empirical validation.

Therefore, this study evaluates the ITCJ analysis using a simulated speech-quality dataset, benchmarking its performance against the CCJ analysis. The study also demonstrates the practical implementation of the approach, adopting a tutorial format with step-by-step guidance on data simulation, prior specification, model estimation, and interpretation using `R`, `Stan`, `cmdstan`, and `brms`. Ultimately, the study equips researchers with practical tools to apply the ITCJ analysis to more complex CJ assessments.

Keywords: tutorial, causal inference, bayesian inference, thustone, comparative judgement

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1. Introduction

Comparative judgment (CJ) has emerged as a valuable methodology for measuring latent traits across diverse fields, including education (Kimbell, 2012; Jones and Inglis, 2015; van Daal et al., 2016; Bartholomew et al., 2018), psychology (Maydeu-Olivares and Böckenholt, 2005), political sciences (ZuccoJr. et al., 2019), linguistics (Boonen et al., 2020), and criminology (Seymour et al., 2022; Seymour and Hernandez, 2025). In CJ studies, judges actively compare pairs of stimuli to determine which stimulus exhibits more of the latent trait of interest (Thurstone, 1927b,a).

The *Classical Comparative Judgment* (CCJ) analysis has become the standard approach for analyzing CJ data across multiple fields, including education, linguistics, and psychology (see, for example, Wu, 2025; Thwaites and Paquot, 2024; Maydeu-Olivares and Böckenholt, 2005). Researchers favor this approach because it provides a convenient and simple method for measuring traits and conducting analyses (Andrich, 1978; Pollitt, 2012b). Three features underpin its simplicity. First, it relies on restricted Thurstone-based models to estimate latent traits. These models facilitate estimation by imposing simplifying assumptions about the traits, judges, and stimuli involved in the CJ assessment under study (Thurstone, 1927a; Bramley, 2008). Second, it pays limited attention to relevant CJ assessment design features that shape the data-generating process, such as the sampling and comparison mechanisms and potential judge biases (Rivera et al., 2025). Third, the approach separates trait estimation from follow-up analysis (Pollitt, 2012b), simplifying the latter by focusing only on mean trait estimates or residuals obtained from the former process (Rivera et al., 2025).

Recent studies, however, question whether the assumptions of these models hold in contemporary CJ applications, and whether the limited consideration of relevant assessment design features, together with the separation of trait estimation and analysis, may bias estimates and reduce the precision of inferences. (Bramley, 2008; Kelly et al., 2022; Rivera et al., 2025).

To address these concerns, Rivera et al. (2025) proposed the *Information-Theoretical Comparative Judgment* (ITCJ) analysis, an umbrella analytical workflow designed to accommodate the complexity often observed in applied CJ settings. The approach leverages causal and Bayesian inference methods to combine Thurstone's core theoretical principles with relevant CJ assessment design features. This integration enables researchers to develop models tailored to the assumed data-generating process of the CJ system under study, thereby reducing reliance on simplifying assumptions. Moreover, by integrating measurement, analysis, and relevant design features within a single framework, the ITCJ analysis facilitates a coherent and flexible approach for trait estimation and statistical inference.

The flexibility of the ITCJ analysis is especially valuable in education, linguistics, and psychology, where CJ data often arise from complex assessment designs (Bramley, 2008; van Daal et al., 2017;

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Kelly et al., 2022) or involve comparisons of complex, heterogeneous stimuli (van Daal et al., 2016; Lesterhuis et al., 2018; Chambers and Cunningham, 2022). Its relevance, however, extends beyond these fields to any discipline using CJ assessments, as the approach allows researchers to model the processes they aim to study rather than those constrained by restrictive statistical models.

1.1. Research goals

The ITCJ analysis holds theoretical promise for producing reliable trait estimates and accurate statistical inferences (Rivera et al., 2025). However, as the authors acknowledge, this promise remains empirically untested. This study addresses this gap through two closely related research goals. First, it *empirically validates the ITCJ analysis* using a simulated speech-quality dataset, assessing the accuracy and reliability of its trait estimates and inferential parameters relative to the standard CCJ analysis.

Second, given the methodological complexity of the ITCJ approach, the study *demonstrates its practical implementation*, aiming to equip researchers with the tools needed to apply the ITCJ analysis to more complex CJ assessments. To this end, the study adopts a tutorial format, providing step-by-step guidance on data simulation, prior specification, model estimation, and interpretation using `R` (R Core Team, 2015), `Stan` (Stan Development Team., 2026b,a), and the interface packages `cmdstanr` (Gabry et al., 2025) and `brms` (Bürkner, 2017, 2018).

The remainder of this manuscript is organized into five sections. Section 2 reviews the standard analytical approach applied to CJ data: the CCJ analysis, and compares it with the ITCJ analysis. Section 3 describes the assumed data-generating process for the simulated dataset, the simulation procedure, the practical implementation of each analytical approach, and the evaluation criteria aligned with the research goals. Section 4 presents the data description and modeling results. Section 5 interprets the findings, outlines future research directions, and considers the study limitations. Finally, Section 6 offers the concluding remarks.

2. A tale of two analytical approaches

Pairwise comparison data are typically analyzed using the standard *CCJ analysis*. However, the *ITCJ analysis* proposed by Rivera and colleagues (2025) offers an alternative approach. This section provides an overview of both.

2.1. The CCJ analysis

The CCJ analysis typically proceeds through two sequential steps, each serving a distinct purpose: first, estimating traits, and second, conducting follow-up analyses (Pollitt, 2012a,b; Jones et al., 2019; Boonen et al., 2020; Chambers and Cunningham, 2022; Bouwer et al., 2023). Trait estimation generally relies on Thurstone-based models and yields two primary outputs: mean trait estimates

and comparison-level residuals. For instance, Boonen et al. (2020) used the Bradley-Terry-Luce (BTL) model (Bradley and Terry, 1952; Luce, 1959) to derive mean latent scores representing children's overall speech quality, together with the corresponding comparison-level residuals.

Researchers then use these two outputs to conduct follow-up analyses. On the one hand, analyses of the mean latent scores fulfill several purposes, including summarizing estimates to aid interpretation, aggregating stimulus-level estimates to the individual level, partitioning variability between and within individuals, and drawing inferences about factors that influence trait scores. For example, Maydeu-Olivares and Böckenholt (2005) applied various factor models to latent means that represent individuals' preferences for compact cars, to provide a more parsimonious representation of these preferences. Similarly, Boonen et al. (2020) applied a multilevel regression model to stimulus mean latent scores to examine whether children's age or hearing status affects their speech quality scores. On the other hand, analyzing the residuals typically seeks to aggregate the remaining variability at the judge level, partition residual variability between and within judges, test for systematic biases, and identify potential misfitting stimuli or judges. For instance, Wu (2025) fitted an analysis of variance (ANOVA) model to judges' infit statistics to examine whether judge expertise influences judgment behavior. The infit statistics–a weighted average of the squared Pearson residuals–potentially indicate the extent to which judges' decisions deviate from expectations relative to those of other judges (Wright and Masters, 1982).

This two-step CCJ analysis has become standard practice in education, linguistics, and psychology (see, e.g., Wu, 2025; Thwaites and Paquot, 2024; Maydeu-Olivares and Böckenholt, 2005). Nevertheless, despite its widespread adoption, this approach has three key limitations: first, it relies on restricted Thurstone-based models to estimate traits; second, it often gives limited consideration to relevant CJ assessment design features; and third, it separates trait estimation from follow-up analysis.

Regarding the first limitation, although restricted Thurstone-based models facilitate trait estimation by imposing simplifying assumptions about traits, judges, and stimuli (Thurstone, 1927a; Bramley, 2008), violations of these assumptions can undermine the reliability and validity of the resulting trait estimates (Perron and Gillespie, 2015; Rivera et al., 2025). This concern is exemplified by Wu (2025), who, under the BTL model, implicitly treats second-language English texts as independent stimuli that are equally informative about individuals' writing performance, despite their heterogeneous and complex nature (Thurstone, 1927b; Andrich, 1978; Bramley, 2008; Kelly et al., 2022). Notably, if the texts differ in informativeness due to this heterogeneous nature, the BTL assumption of equal dispersions may be violated, potentially leading to an overestimation of the trait reliability and misleading conclusions about differences between stimuli (McElreath, 2020; Wu et al., 2022). Moreover, when text characteristics unevenly influence judges' perceptions, systematic biases may arise. Such biases can induce dependencies among stimuli, further contributing to an over- or

underestimation of the trait reliability (van der Linden, 2017; Ackerman, 1989; Hoyle, 2023).

Concerning the second limitation, while giving limited attention to specific CJ assessment design features may also simplify trait estimation or follow-up analyses, Rivera and colleagues (2025) argued that neglecting elements that define the CJ data-generating process can obscure their broader role within the assessment system. The authors illustrated this concern by examining the sampling and comparison mechanisms. They showed that explicit inclusion of these mechanisms in the conceptual CJ model reveal their role as sources of missing data. They further argued that clarifying their function supports a more comprehensive evaluation of their impact on trait estimation, statistical inference, and the design of more complex assessment setups.

Lastly, concerning the third limitation, although separating trait estimation from follow-up analysis can simplify the latter, focusing solely on mean trait estimates or residuals obtained from the estimation stage inadvertently treats these outputs as fixed values rather than as uncertain parameter estimates (Rivera et al., 2025). This separation thus prevents the propagation of uncertainty from the estimation stage to the follow-up analyses (Pritikin, 2020), which can bias parameter estimates and reduce their precision. The direction and magnitude of these biases are often difficult to predict. Effects may be attenuated, inflated, or remain unchanged depending on the level of uncertainty in the scores and the true effects under investigation (McElreath, 2020; Kline, 2023; Hoyle, 2023). In addition, reduced precision lowers statistical power and increases the risk of Type I and Type II errors (McElreath, 2020).

Several studies have made valuable attempts to incorporate specific features that address some of the limitations discussed above. However, most studies propose models that address only one or two issues at a time. As a result, other relevant components of the CJ process remain unrepresented, which may lead to model misspecification and produce biased or imprecise parameter estimates (Everitt and Skrondal, 2010). For instance, Seymour and Hernandez (2025) extended the BTL model to incorporate spatially correlated stimuli and tied outcomes, applying it to measure honour-based abuse. While these features captured important aspects of the CJ data, the analysis did not explicitly model other potentially relevant factors, such as the sampling mechanism associated with the convenience sample, possible judge biases and judge-specific effects, or the hierarchical structures such as wards nested within counties. Similar patterns appear in Pritikin (2020), Schramm and Batchelder (2021), Seymour et al. (2022), and Gray et al. (2024), where models typically address one or two key design features or sources of heterogeneity at a time.

Notably, the ITCJ analysis directly addresses these all of the limitations by providing a unified and systematic framework for analyzing CJ data (Rivera et al., 2025). Specifically, it allows researchers to model the processes of substantive interest while propagating uncertainty across components, rather than relying on restrictive statistical model specifications.

2.2. The ITCJ analysis

The ITCJ analysis begins by explicitly representing the relationships among observed judgments, stimulus traits, individual traits, judge biases, and the sampling and comparison mechanisms using a general *Directed Acyclic Graph (DAG)* and a corresponding *Structural Causal Model (SCM)* (Morgan and Winship, 2014; Gross et al., 2018; Neal, 2020). This general DAG and SCM establish a coherent theoretical foundation for analysis of CJ data. Next, the approach adapts the general SCM and DAG to the assumed data-generating process of the CJ system under study. It then derives one or more bespoke *probabilistic* and *statistical* models tailored to that system. Finally, it uses one or more of these models to estimate, summarize, or conduct statistical inference on the traits. Figure 1 illustrates an example of a general DAG structure.

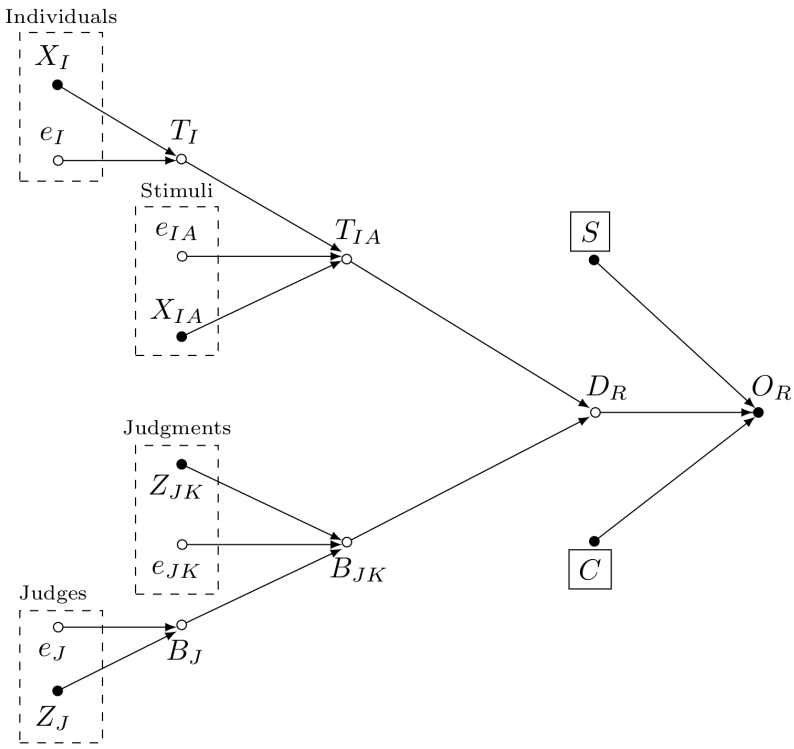

Figure 1: ITCJ analysis: DAG

In this representation, O_R denotes the observed judgment outcome vector, and D_R represents the discriminal difference vector. T_{IA} captures the vector of stimulus traits, and T_I represents the vector of individual traits. The vector of judgment biases is represented by B_{JK}, while the vector of judges' biases by B_J. Covariates at the stimulus and individual levels appear as X_{IA} and X_I, respectively; while Z_{JK} and Z_J represent covariates at the judgment and judge levels. The error terms (e_{IA}, e_I, e_{JK}, e_J) capture residual variation at each level.

For example, Rivera et al. (2025) exemplified a CJ assessment designed to evaluate the impact of different teaching methods on students' writing ability. In this case, the observed outcome was

binary, so the model assumed O_R followed a Bernoulli distribution. The discriminal difference (D_R) was determined by the texts' discriminal processes (T_{IA}) and the judgment biases (B_{JK}). Student-level variables X_I, such as teaching method, were included, whereas text-level variables X_{IA} (e.g., text length) were not gathered. Similarly, judge-level variables Z_J, like judgment expertise, were incorporated, while judgment-level variables Z_{JK} (e.g., number of judgments per judge) were absent. Finally, the probabilistic assumptions for the idiosyncratic errors (e_I, e_{IA}, e_J, e_{JK}) were specifically made to resolve indeterminacies in *location*, *orientation*, and *scale* for the variables T_I, T_{IA}, B_J, B_{JK}, as required in latent variable models (Depaoli, 2021; de Ayala, 2009).

The DAG also represents the sampling and comparison mechanisms as the vectors S and C, two conditioned variables that determine how population outcomes become "observed" (sampled) outcomes. Importantly, the DAG depicts S and C as independent from all other variables by showing no arrows pointing into them. Regarding S, this indicates that the DAG applies to *Simple Random Sampling (SRSg)* designs, where each repeated judgment, judge, stimulus, and individual has an equal probability of inclusion within their respective groups (Lawson, 2015). Regarding C, the DAG applies to *Random Allocation Comparative Designs* (Bramley, 2015) or *Incomplete Block Designs* (Lawson, 2015), where every repeated judgment has an equal chance of being included in the sample.

Researchers can then translate this DAG and SCM representations into probabilistic forms that express the joint distribution of a complex CJ system as a product of simpler *conditional probability distributions (CPDs)* (Pearl et al., 2016; Neal, 2020; see also Rivera et al., 2025, section 5), as illustrated in Figure 2.

$$
\begin{aligned}
O_R &:= f_O(D_R, S, C) \\
D_R &:= f_D(T_{IA}, B_{JK}) \\
T_{IA} &:= f_T(T_I, X_{IA}, e_{IA}) \\
T_I &:= f_T(X_I, e_I) \\
B_{JK} &:= f_B(B_J, Z_{JK}, e_{JK}) \\
B_J &:= f_B(Z_J, e_J) \\
\\
e_I &\perp \{e_J, e_{IA}, e_{JK}\} \\
e_J &\perp \{e_{IA}, e_{JK}\} \\
e_{IA} &\perp e_{JK}
\end{aligned}
$$

(a) SCM

$$
\begin{aligned}
&P(O_R \mid D_R, S, C) \\
&P(D_R \mid T_{IA}, B_{JK}) \\
&P(T_{IA} \mid T_I, X_{IA}, e_{IA}) \\
&P(T_I \mid X_I, e_I) \\
&P(B_{JK} \mid B_J, Z_{JK}, e_{JK}) \\
&P(B_J \mid Z_J, e_J) \\
\\
&P(e_I)P(e_{IA})P(e_J)P(e_{JK})
\end{aligned}
$$

(b) Probabilistic model

Figure 2: ITCJ analysis. SCM (left) and probabilistic (right) representations for DAG in Figure 1.

Critically, the approach allows tailoring this general structure to a specific CJ context, enabling development of parsimonious models that match the assumed data-generating process without imposing unnecessary constraints.

Lastly, researchers can derive one or more *bespoke* statistical models tailored to the CJ system of interest, as demonstrated in Rivera et al. (2025). At this stage, the ITCJ analysis differs fundamentally from the CBTL approach in how it handles parameter estimation and inference. Rather than fitting multiple separate models, the approach simultaneously estimates all parameters within a single coherent framework using *Bayesian inference*. This joint estimation accounts for uncertainty at all levels and enables direct inference about quantities of interest without relying on post-hoc procedures (McElreath, 2020).

3. Methods

3.1. Step 1, from Theory to Design: Data-generating assumptions

3.2. Step 2, from Design to Data: Data simulation

3.3. Step 5, from Estimator and Sample to Estimate(s): The analysis aproaches

3.3.1. The CCJ analysis

3.3.2. The ITCJ analysis

3.3.2.1. Model 1.

3.3.2.2. Model 2.

3.3.2.3. Model 3.

3.3.2.4. Model 4.

3.3.2.5. Model 5.

3.3.2.6. Model 6.

3.4. Step 6, from Estimate(s) to Diagnostics and Posterior predictives: The evaluation criteria

4. Results

4.1. Data description

4.2. Data modeling

4.2.1. The CCJ analysis

4.2.2. The ITCJ analysis

4.2.2.1. Model 1.

4.2.2.2. Model 2.

4.2.2.3. Model 3.

4.2.2.4. Model 4.

4.2.2.5. Model 5.

4.2.2.6. Model 6.

4.2.2.7. Model comparison.

5. Discussion

5.1. Future research directions

5.2. Study limitations

6. Conclusion

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Licence: All the code that is original to this study and not attributed to any other authors is copyrighted by Jose Manuel Rivera Espejo and released under the new BSD-3-Clause license.

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7. Appendix

7.1. Appendix A: Stationarity, converge and mixing

7.2. Appendix B: Misfit observations

7.3. Appendix C: Sample size calculations

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